

Sanath N Upadhyaya

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<https://sanathnu.github.io> | <https://github.com/sanathNU>

PROFESSIONAL SUMMARY

Software engineer with experience building backend components for large-scale distributed database systems and cloud infrastructure. Focused on scalability, fault tolerance, and observability, with hands-on experience in Kubernetes, CI/CD automation, and secure service design.

EDUCATION

Master of Science in Computer Science

Northeastern University, Boston, MA

Aug 2025 - Present

GPA: 4.0

Coursework: Programming Design Paradigm, Reinforcement Learning, Algorithms, Database Systems

Bachelor of Engineering (Hons) in Electronics and Communication Engineering

BMS College of Engineering, Bengaluru, India

2019 - 2023

GPA: 9.15 / 10

Coursework: Operating Systems, Machine Learning, Internet of Things, Number Theory

WORK EXPERIENCE

Software Development Engineer Intern | Amazon Web Services, Palo Alto, CA

May 2026 - Present

- Architecting and developing an automated integration testing framework for Aurora MySQL from the ground up, generating and validating third-party workloads against distributed database clusters.
- Implementing differential testing pipelines using oracle databases to identify correctness discrepancies across query execution paths.
- Designed custom agent orchestration workflows to automate test generation, execution, and validation, significantly reducing manual effort in large-scale compatibility testing.
- Contributing to backend components for Aurora Open Source Engines, focusing on scalability and fault tolerance.
- Integrated automated validation workflows into existing testing pipelines to expand compatibility coverage across third-party workloads.

Cloud Developer | Hewlett Packard Enterprise, Bengaluru, India

Aug 2023 - Aug 2025

- Implemented OAuth2- and Keycloak-based authentication pipelines, migrating legacy LDAP systems to centralized identity platforms using GreenLake IAM and Okta.
- Designed zero-trust microservice authentication using Istio and SPIRE for secure service-to-service communication.
- Integrated backend services and automation into CI/CD pipelines (Jenkins, GitHub Actions), improving deployment reliability through automated testing.
- Developed storage integrations using MinIO and Kubernetes storage operators to support cloud-native workloads.
- Implemented automated unit and integration testing using pytest and Selenium, increasing release confidence and reducing deployment regressions across cloud services.
- Led development of Private Cloud AI Essentials (PCAI), enabling rapid access to AI hardware resources and reducing deployment time by 40%.

Software Engineering Intern | Hewlett Packard Enterprise, Bengaluru, India

Jan 2023 - July 2023

- Implemented secure access for Ezmeral Runtime Engine using Keycloak and OAuth2-proxy; enabled OpenShift support for HPE Unified Analytics and optimized Prometheus performance.
- Automated Red Hat OpenShift cluster installation using Python and REST APIs, reducing deployment time by 30%.

PROJECTS

Akshara Mantapa: Infinite Kannada Text Library | [Github](#) | [Web-Demo](#)

Rust, WebAssembly, SvelteKit

- Built a distributed text-generation and retrieval system in Rust and WebAssembly supporting deterministic indexing and search across Kannada, Telugu, and Tamil, spanning over 130k+ valid Indic grapheme clusters through a generic Brahmic script abstraction.
- Developed Unicode-aware indexing and segmentation pipelines to ensure correct search and retrieval across complex Indic grapheme clusters under adversarial inputs.

Post-Quantum & ZK Proof Benchmarking Framework | [Github](#)

Rust, ml-kem, arkworks, Docker, Kubernetes

- Designed Rust microservices benchmarking ML-KEM and Groth16 under distributed workloads, identifying a ~17x performance gap and evaluating scalability using Kubernetes.

End-to-End Encryption Engine with Double Ratchet | [Github](#)

Java, Java Cryptography Extension (JCE),

- Implemented a Double Ratchet-based encrypted messaging system achieving forward secrecy and post-compromise security.

TECHNICAL SKILLS

Languages : Python, Go, Rust, C++, Java, Bash

Cloud Platforms: AWS (EC2, EKS, RDS, S3), GCP (GKE, Cloud Run), Azure

Distributed Systems and Infrastructure: PostgreSQL, MySQL, Redis, Apache Kafka, RabbitMQ

DevOps and Security: Jenkins, Github Actions, Prometheus, Ansible, Terraform, Keycloak

AI Systems & Automation: Multi-agent orchestration, Differential testing, LLM application development, RAG pipelines

Certifications : Certified Kubernetes Administrator (CKA), Certified Kubernetes Security Specialist (CKS)